- PDU - protocol data unit
- protocol
- peer process ⇝ protocol on same level = peers
- PDU at layer 2, 3, 4

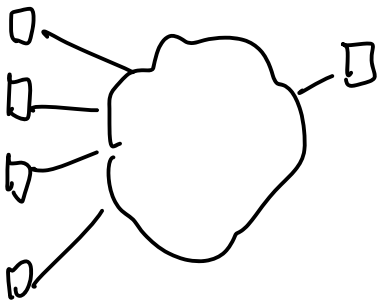
- layer 3: router
- layer 4: error correction, end-end bitstream

Client/Server

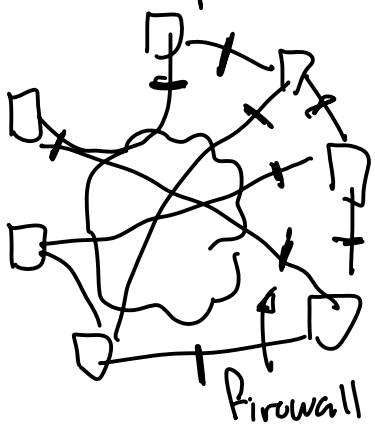
Peer-to-peer

- Local Area Network (LAN)
- Wide Area Network (WAN) ~ anything that's not us
- Last Mile ⇝ last connection from ISP to house
- Internet Service Provider (ISP) ~ part of internet
  - Comcast
  - AT&T

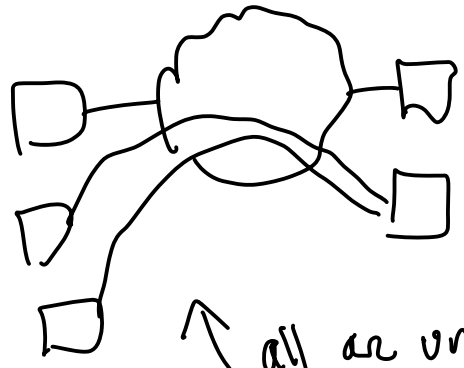
Client_server



peer-to-peer



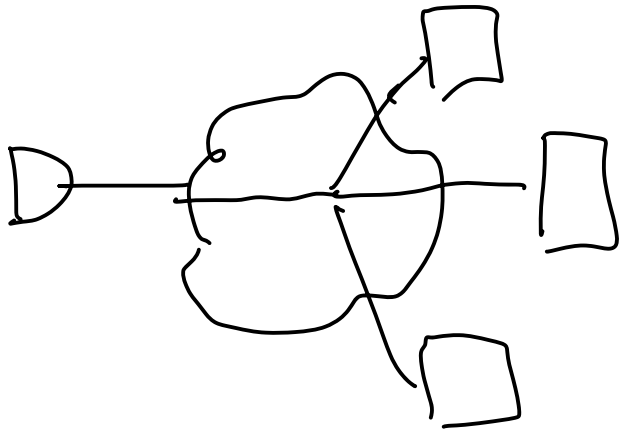
Unicast: 1-1



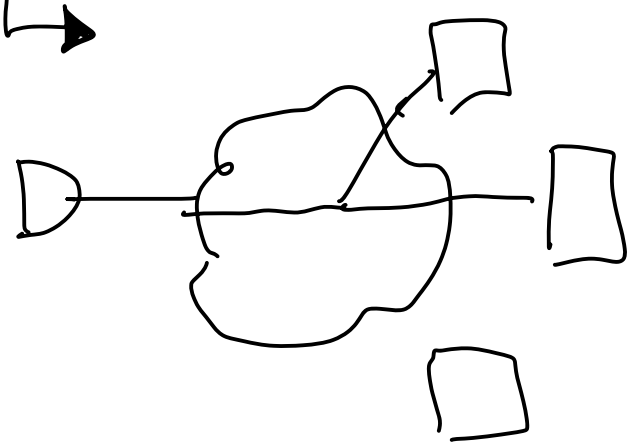
all are unicast

Broadcast Connection

talk / connected to all others



Multi Cast



TCP

Connection Oriented (Reliable)

Client-server

1- setup (I want to setup)

2 - use

3 - teardown

Connectionless (UDP) unreliable

- use
  - no setup
  - no teardown

Throughput ETH, etc are connectionless.

→ bits/sec or bytes/sec

Gigabit & Gigabyte

Kb = ~~2^{10}b~~ = 10^3 = 1000

Mbps = 10^6 = 100,000

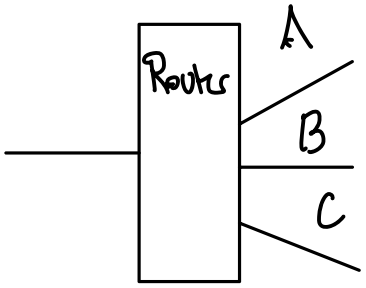
NIC = network interface card → not really a card anymore

- hardware + software

Packet Switching & Packets:

Layer 3 datagram = packet

Addressing Information

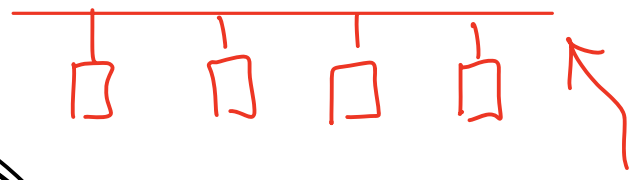


layer 4 doesn't have addressing information

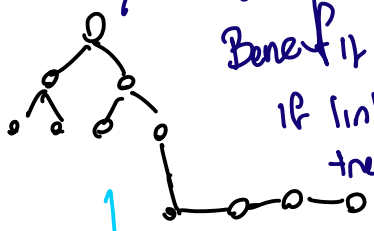
Router Switching Table

Networking Topologies

- Bus / Broadcast → all devices see the traffic (security issue)
  - only one can talk at a time



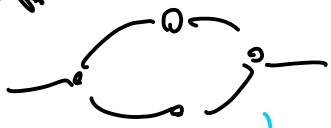
(no cycles / loops) - Tree / Hierarchical



Routing Benefit, but if links break, tree breaks

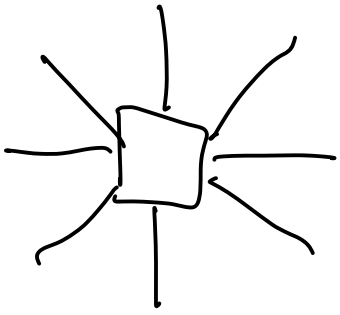
↳ easy to address
- no redundancy

- Ring



- good, redundancy

- Star

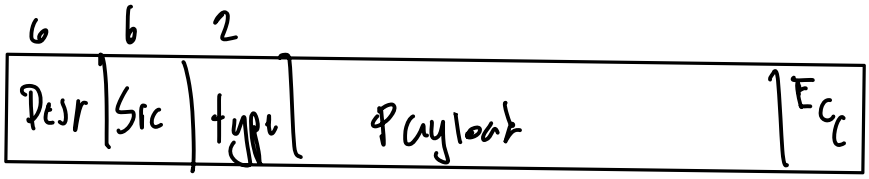


- Ad-Hoc (mesh)

- Physical vs. Logical

WiFi encrypts network traffic

Lab tomorrow



Eth header = 14 bytes

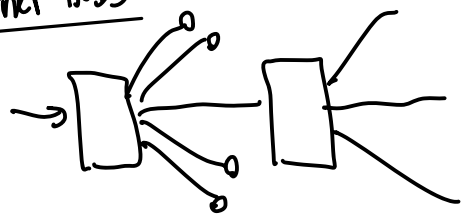
Hardware address of NIC

MAC: 2F : 34 : da : 21 : 5e : 01

Address Learning:

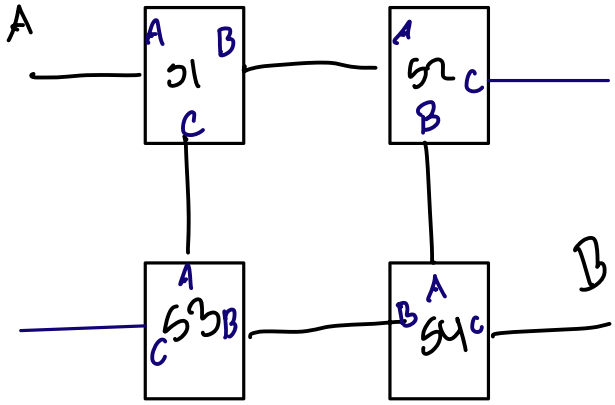
2 phases
1. learning

Ethernet Hubs



Ethernet Switches

Switch creates MAC address table



- switch doesn't want to broadcast
- MAC address generated passively

Dest	Interface

Dest MAC	Next_Hop

Switch Learning:

- Learns source MAC address and interface it came in on (interface A)

- PDU showed up at SW, SW learned MAC and put it on table.
  - if destination MAC is not in the table, it'll broadcast frames and its internet

PC1 sends frame
- switch learns MAC 1
- we pickup source MAC to learn, not destination

Port Spanning: duplicate traffic on another interface

STP (spanning tree protocol)